

Moral Hazard with Dual Risk-Averse Agent*

Yusuke Yamaguchi [†]

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Abstract

This paper studies a stylized moral hazard problem with a single agent protected by the limited liability constraint and is *dual risk-averse* in the sense of [Yaari \(1987\)](#). I consider the principal's cost-minimizing problem to induce a given action. Under some assumption on risk-aversion, which can be interpreted as the agent overweighting the probability of the worst outcome, and the strict version of the monotone likelihood ratio property, the solution to a relaxed problem where the agent's incentive compatibility constraint is replaced by its first-order condition is *debt contract*—the wage is constant up to some level of output, and then grows with the same speed as output afterwards. I also show that the first-order approach is valid if the outcome distribution is (in proportionate terms) less convex in action than the disutility of action, which is implied by [Rogerson \(1985\)](#)'s Convexity of Distribution Function Condition and convex disutility.

1 Introduction

Since the seminal work of [Holmström \(1979\)](#), a large literature on moral hazard has studied how a principal can incentivize an agent to take desirable actions when the agent's action is not contractible. A persistent critique of the theory is that it often predicts optimal contracts that are complex and highly sensitive to the underlying distribution of outcomes, in contrast to the simple contracts frequently observed in practice.

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[†]Institute of Social and Economic Research (ISER), The University of Osaka, 6-6-1 Mihogaoka, Ibaraki, Osaka 567-0047, Japan. Email: yamaguchi@iser.osaka-u.ac.jp

This paper offers an alternative rationale for such simple linear contracts; dual risk-aversion implies the optimality of debt contracts. I consider a stylized moral hazard model with a single agent who is protected by the limited liability constraint and *dual risk-averse* in the sense of Yaari (1987); that is, he overweights the probability of “bad” outcomes when making decisions. In particular, I consider one of the simplest (dual) risk-aversion where the agent behaves as if he believes that the worst outcome is realized with probability $(1 - k)$ and that outcome follows some distribution with probability k , where $k \in (0, 1)$. If I restrict the class of admissible contracts to those monotonically increasing and whose slope is at most 1, under the strict version of the monotone likelihood ratio property, Pontryagin’s maximum principle characterizes the solution to the principal’s problem. For each inducible action, the cost-minimizing contract for the principal is *debt contract*; the wage is constant up to some level of output, and then grows with the same speed as output afterwards (Proposition 1). I also show that the solution to the relaxed problem is indeed the solution to the original problem if the outcome distribution is (in proportionate terms) less convex in action than the disutility of action (Proposition 2).

This paper contributes to the literature on moral hazard by providing a novel rationale—dual risk-aversion—for the simple contracts observed in practice. Proposition 1 shows the optimality of debt contracts under the monotonicity and the limited liability constraints, which echoes the finding of Innes (1990), who also shows the optimality of debt contracts under similar constraints. Holmström and Milgrom (1987) also shows the optimality of linear contracts in certain environments. This paper also contributes to the literature that explores sufficient conditions for validating the so-called first-order approach (FOA)—replacing the agent’s incentive compatibility constraint by its first-order condition. The seminal work of Rogerson (1985) shows that the FOA is valid if the distribution of outcome is convex in action (Convexity of Distribution Function Condition). Recently, Chade and Swinkels (2020) shows a more general sufficient condition that validates the FOA. The proof of Proposition 2 follows almost the same steps as in Chade and Swinkels (2020). As in Chade and Swinkels (2020), the condition in Proposition 2 is implied by Convexity of Distribution Function Condition and convex disutility.

The remainder of the paper is organized as follows. Section 2 introduces the model. I present the results of the paper in Section 3, and Section 4 concludes. The proof of Proposition 1 is

relegated to Appendix B.

2 Model

Environment: I consider a stylized moral hazard environment with a single principal and a single agent. The agent takes a hidden action $a \in [0, \bar{a}]$ at a disutility $\psi(a)$, which is strictly increasing and C^2 with $\psi(0) = 0$. I assume that ψ satisfies the Inada conditions: $\psi'(0) = 0$ and $\psi'(\bar{a}) = +\infty$. The agent's action affects the distribution of outcome x , which takes a value in $[\underline{x}, \bar{x}]$. Given action a , the outcome x has a distribution $F(x | a)$ with everywhere positive density $f(x | a)$. I assume that F and f are C^2 and that the agent's action produces a first-order stochastic dominant shift on the outcome; $F_a(x | a) < 0$ for all $x \in (\underline{x}, \bar{x})$ and all a .¹

Preferences: The principal offers a bounded wage schedule $w: [\underline{x}, \bar{x}] \rightarrow [-m, M]$ to the agent, where $m, M \in \mathbb{R}_+$ and M is sufficiently large. She is risk-neutral and maximizes her expected net benefit $\mathbb{E}[x - w(x)]$. The agent is *dual risk-averse* in the sense of Yaari (1987): given the distribution of wage $\Gamma^w(\cdot | a)$ induced by wage schedule w and action a , his gross utility is given by

$$\int_{-m}^M \phi(1 - \Gamma^w(v | a)) dv, \quad (2.1)$$

where distortion function $\phi: [0, 1] \rightarrow [0, 1]$ is defined as

$$\phi(p) = \begin{cases} kp & \text{if } 0 \leq p < 1 \\ 1 & \text{if } p = 1 \end{cases}$$

for some $k \in (0, 1)$.

As I discuss in Section 3, this distortion ϕ can be interpreted as the agent's pessimistic belief under which he believes that the worst outcome $\underline{x}$ is realized with probability $1 - k$ and that outcome follows F with probability k .

Remark 1. Yaari (1987) characterizes the dual representation for general continuous and nondecreasing ϕ satisfying $\phi(0) = 0$ and $\phi(1) = 1$. He also characterizes risk-aversion by the

¹Note that $F_a(\underline{x} | a) = F_a(\bar{x} | a) = 0$ for all a since the support of the outcome is fixed.

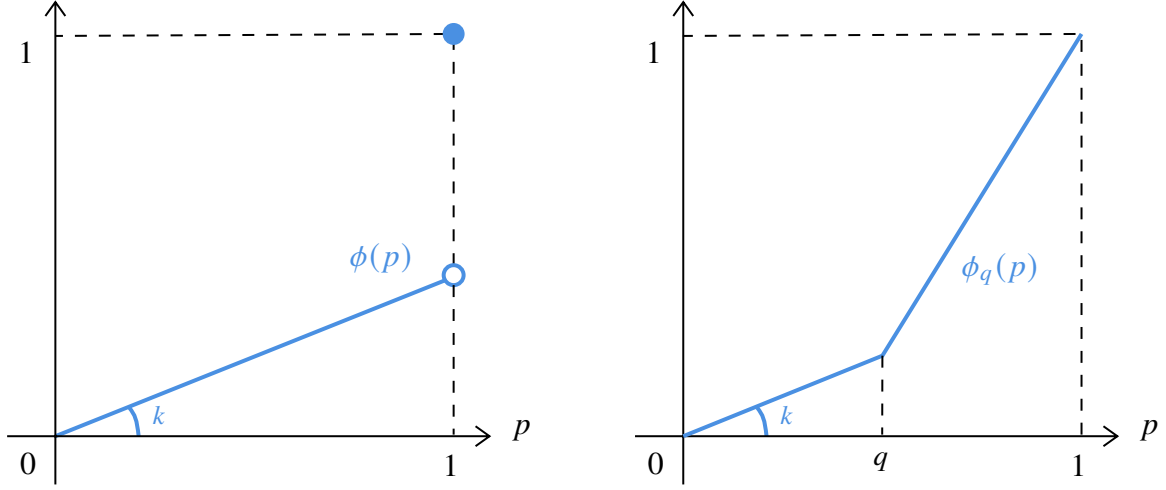


Figure 2.1: Distortion function ϕ can be interpreted as the limit of ϕ_q where q goes to 1.

convexity of ϕ . The present ϕ can be interpreted as the limit of the convex and piecewise-linear function that has a slope k on $[0, q)$ and extends from there to $(1, 1)$. See Figure 2.1.

Admissible wage schedules: I assume $m = 0$ so that the agent is protected by the limited liability constraint. Following [Poblete and Spulber \(2012\)](#), I restrict attention to wage schedules that satisfy the following assumptions.

Assumption 1 (Monotonicity). $w(x)$ is nondecreasing.

Assumption 2 (Bounded slope). $x - w(x)$ is nondecreasing.

Assumption 1 prevents the agent from sabotaging the task while Assumption 2 prevents the principal from doing so after the action has been chosen. The latter also prevents the agent from inflating the outcome by borrowing money to increase his returns. Assumptions 1 and 2 imply $w(x) \geq w(x')$ and $x - w(x) \geq x' - w(x')$ for $x \geq x'$. By taking the limit, this implies that w is almost everywhere differentiable and Lipschitz continuous with

$$0 \leq w'(x) \leq 1 \quad \text{for almost every } x. \quad (\text{Lipschitz})$$

3 Cost-minimizing wage schedules

Suppose that the principal wants to induce action $a > 0$. Given Assumption 1, the distribution of wage is given by $\Gamma^w(v | a) = F(w^{-1}(v) | a)$, where w^{-1} is the generalized inverse of w :

$$w^{-1}(v) = \sup\{x : w(x) \leq v\}.$$

The agent's gross utility (2.1) can be expressed as

$$\begin{aligned} \int_0^M \phi\left(1 - F\left(w^{-1}(v) | a\right)\right) dv &= w(\underline{x}) + \int_{\underline{x}}^{\bar{x}} \phi(1 - F(x | a)) w'(x) dx \\ &= w(\underline{x}) + \int_{\underline{x}}^{\bar{x}} k[1 - F(x | a)] w'(x) dx, \end{aligned}$$

where the first equality follows from the change of variable $x = w^{-1}(v)$. Using integration by parts, this can be further transformed into

$$(1 - k)w(\underline{x}) + k \int_{\underline{x}}^{\bar{x}} f(x | a)w(x) dx. \quad (3.1)$$

That is, the agent behaves as if he believes that the worst outcome $\underline{x}$ is realized with probability $1 - k$ and that outcome follows F with probability k . In this sense, the agent overweights the probability of the bad (indeed, the worst) outcome in making his decision, which is consistent with Yaari (1987)'s interpretation of dual risk-aversion.

3.1 Principal's (relaxed) problem

For the agent to accept wage schedule w and choose action a , his net utility $U(w, a)$ must be greater than or equal to his reservation utility U_0 :

$$U(w, a) \equiv w(\underline{x}) + \int_{\underline{x}}^{\bar{x}} k[1 - F(x | a)] w'(x) dx - \psi(a) \geq U_0. \quad (\text{PC})$$

Given w , he chooses action a if it maximizes $U(w, \tilde{a})$:

$$U(w, a) \geq U(w, \tilde{a}) \quad \text{for all } \tilde{a} \in [0, \bar{a}]. \quad (\text{IC})$$

Its first-order condition $U_a(w, a) = 0$ is equivalent to

$$\int_{\underline{x}}^{\bar{x}} k[-F_a(x | a)] w'(x) dx = \psi'(a). \quad (\text{FOA})$$

Since w is nondecreasing by Assumption 1, the limited liability constraint is reduced to

$$w(\underline{x}) \geq 0. \quad (\text{LL})$$

The principal minimizes the expected payment $\int_{\underline{x}}^{\bar{x}} w(x) f(x | a) dx$ to the agent under the constraints (PC), (IC), (LL), and (Lipschitz). To characterize the solution to this problem, I first consider a relaxed problem where (IC) is replaced by (FOA). Then, I derive a sufficient condition under which the solution to the relaxed problem indeed satisfies (IC); that is, the FOA is valid. The relaxed problem can be formulated as follows:

$$\begin{aligned} \min \quad & \left[\int_{\underline{x}}^{\bar{x}} w(x) f(x | a) dx \right] \\ \text{subject to} \quad & w(\underline{x}) + \int_{\underline{x}}^{\bar{x}} k[1 - F(x | a)] w'(x) dx - \psi(a) \geq U_0 \quad (\text{PC}) \\ & \int_{\underline{x}}^{\bar{x}} k[-F_a(x | a)] w'(x) dx = \psi'(a) \quad (\text{FOA}) \\ & w(\underline{x}) \geq 0 \quad (\text{LL}) \\ & 0 \leq w'(x) \leq 1 \quad \text{for a.e. } x \quad (\text{Lipschitz}) \end{aligned}$$

Given (Lipschitz), the constraint (FOA) can be satisfied only for those actions a satisfying

$$\int_{\underline{x}}^{\bar{x}} k[-F_a(x | a)] dx \geq \psi'(a). \quad (3.2)$$

In the remainder of the paper, I restrict attention to those actions satisfying (3.2) with a strict inequality:²

$$A^* = \left\{ a \in (0, \bar{a}) : \int_{\underline{x}}^{\bar{x}} k[-F_a(x | a)] dx > \psi'(a) \right\}$$

²For action a such that (3.2) holds with equality, the only way to induce action a is to make the agent a residual claimant; that is, $w'(x) = 1$ for all x .

To make the problem amenable to the control-theoretic technique, define α and β by

$$\begin{aligned}\alpha(x) &= w(\underline{x}) + \int_{\underline{x}}^x k[1 - F(\tilde{x} | a)] w'(\tilde{x}) d\tilde{x}, & \alpha'(x) &= k[1 - F(x | a)] w'(x), \\ \beta(x) &= \int_{\underline{x}}^x k[-F_a(\tilde{x} | a)] w'(\tilde{x}) d\tilde{x}, & \beta'(x) &= k[-F_a(x | a)] w'(x).\end{aligned}$$

Then, the problem can be expressed as an optimal control problem with states (w, α, β) and a control w' :

$$\begin{aligned}\min \quad & \left[\int_{\underline{x}}^{\bar{x}} w(x) f(x | a) dx \right] \\ \text{subject to} \quad & w'(x) = u(x) \quad \text{for a.e. } x \\ & \alpha'(x) = k[1 - F(x | a)] u(x), \quad \beta'(x) = k[-F_a(x | a)] u(x) \quad (\text{P}) \\ & w(\underline{x}) \geq 0, \quad w(\bar{x}) \text{ free} \\ & \alpha(\underline{x}) = w(\underline{x}), \quad \alpha(\bar{x}) - \psi(a) \geq U_0, \quad \beta(\underline{x}) = 0, \quad \beta(\bar{x}) = \psi'(a) \\ & 0 \leq u(x) \leq 1 \quad \text{for all } x \in [\underline{x}, \bar{x}].\end{aligned}$$

Lastly, I define a class of simple contracts, debt contracts:³

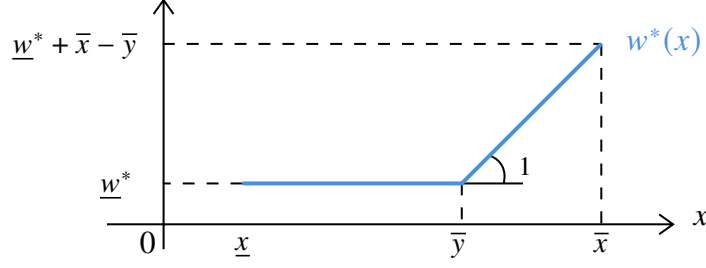
Definition 1. Wage schedule w is a *debt contract* if there exist y such that $w'(x) = 0$ if $x \in [\underline{x}, y)$ and $w'(x) = 1$ if $x \in (y, \bar{x}]$.

I show that, under the strict version of the monotone likelihood ratio property, the solution to (P) for each $a \in A^*$ is a debt contract.

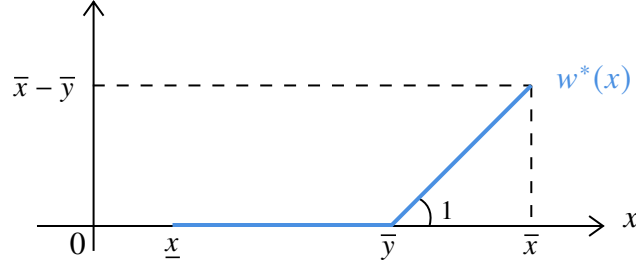
Assumption 3 (Strict MLRP). For each a , $\frac{\partial}{\partial x} \left(\frac{f_a(x|a)}{f(x|a)} \right) > 0$ for all x .

Proposition 1. Assume F satisfies Strict MLRP. For each $a \in A^*$, the cost-minimizing wage schedule w^* associated with the solution u^* to the relaxed problem (P) is debt contract. Specifically, let $\underline{w}^* \equiv U_0 + \psi(a) - \int_{\bar{y}}^{\bar{x}} k[1 - F(x | a)] dx$, where $\bar{y}$ is the unique value satisfying $\int_{\bar{y}}^{\bar{x}} k[-F_a(x | a)] dx = \psi'(a)$.

³In Appendix A, I define *debt and capped bonus contract*, which includes debt contract as a special case.



(i) When $\underline{w}^* > 0$



(ii) When $\underline{w}^* \leq 0$

Figure 3.1: The cost-minimizing wage schedule w^* is always debt contract.

1. If $\underline{w}^* > 0$,

$$w^*(x) = \begin{cases} \underline{w}^* & \text{if } x \in [\underline{x}, \bar{y}] \\ \underline{w}^* + x - \bar{y} & \text{if } x \in (\bar{y}, \bar{x}] \end{cases}$$

2. If $\underline{w}^* \leq 0$,

$$w^*(x) = \begin{cases} 0 & \text{if } x \in [\underline{x}, \bar{y}] \\ x - \bar{y} & \text{if } x \in (\bar{y}, \bar{x}] \end{cases}$$

Proof. See Appendix B. □

The solution w^* is depicted in Figure 3.1. As I discussed at the beginning of this section, the agent behaves as if he is pessimistic, overweighting the probability of the worst outcome. Although he has such pessimism, the agent can be considered as “almost” risk-neutral because the second term in (3.1) is the (conditional) expected wage $\mathbb{E}[w(x)|a]$ scaled by k . He is (dual) risk-averse only in the sense implied by his pessimism; that is, by the presence of the first term $(1 - k)w(\underline{x})$.⁴ In this view, Proposition 1 is consistent with Innes (1990), who has shown the

⁴Yaari (1987) discusses that risk-neutrality is characterized by ϕ being the identity function, which is—of course—linear. Since ϕ in the present paper is linear except at one point, this also indicates that the agent is almost risk-neutral.

optimality of debt contracts under risk-neutrality and the limited liability and monotonicity constraints.

3.2 The validity of the first-order approach

I now turn to the validity of the FOA. Using the technique in [Chade and Swinkels \(2020\)](#), I derive a sufficient condition under which the agent's net utility is strictly quasiconcave in action; thus, the (FOA) characterizes the optimal choice of action. As in [Chade and Swinkels \(2020\)](#), the concavities (in proportionate terms) of outcome distribution F and disutility ψ are the key to validating the FOA.

Proposition 2. *Under Strict MLRP, the first-order approach is valid if*

$$\frac{F_{aa}(x | a)}{F_a(x | a)} < \frac{\psi''(a)}{\psi'(a)} \quad \text{for all } x \in (\bar{y}, \bar{x}). \quad (3.3)$$

Proof. The proof follows almost the same steps as [Chade and Swinkels \(2020, Proposition 5\)](#).

Recall that the agent's net utility from wage schedule w and action $\tilde{a}$ is given by

$$U(w, \tilde{a}) = w(\underline{x}) + \int_{\underline{x}}^{\bar{x}} k[1 - F(x | \tilde{a})] w'(x) dx - \psi(\tilde{a}).$$

Its first- and second-order derivatives with respect to action are

$$\begin{aligned} U_a(w, \tilde{a}) &= \int_{\underline{x}}^{\bar{x}} k[-F_a(x | a)] w'(x) dx - \psi'(a) \\ U_{aa}(w, \tilde{a}) &= \int_{\underline{x}}^{\bar{x}} k[-F_{aa}(x | a)] w'(x) dx - \psi''(a), \end{aligned}$$

If $U_a(w, a) = 0$,

$$\psi''(a) = \frac{\psi''(a)}{\psi'(a)} \psi'(a) = \int_{\underline{x}}^{\bar{x}} \frac{\psi''(a)}{\psi'(a)} k[-F_a(x | a)] w'(x) dx.$$

Using this equation, $U_{aa}(w, a)$ can be written as

$$U_{aa}(w, a) = -k \int_{\underline{x}}^{\bar{x}} \left[F_{aa}(x | a) - \frac{\psi''(a)}{\psi'(a)} F_a(x | a) \right] w'(x) dx. \quad (3.4)$$

Note that $U_{aa}(w, a) < 0$ suffices for $U(w, \cdot)$ to be strictly quasiconcave. For w^* in Proposition 1, this condition is reduced to

$$\int_{\bar{y}}^{\bar{x}} \left[F_{aa}(x | a) - \frac{\psi''(a)}{\psi'(a)} F_a(x | a) \right] dx > 0.$$

This holds if

$$\frac{F_{aa}(x | a)}{F_a(x | a)} < \frac{\psi''(a)}{\psi'(a)} \quad \text{for all } x \in (\bar{y}, \bar{x}).$$

□

The condition (3.3) requires that for all $x \in (\bar{y}, \bar{x})$, the outcome distribution F is (in proportionate terms) less convex in action than the disutility ψ . In particular, (3.3) is satisfied if both F and ψ are convex in action, as in Rogerson (1985).

4 Conclusion

This paper has explored the implications of dual risk-aversion in a standard moral hazard setting with monotonicity and limited liability constraints. Under a specific form of risk-aversion where the agent overweights the probability of the worst outcome, the cost-minimizing contract for the principal takes a particularly simple form, a debt contract. The result echoes the finding of Innes (1990), who has shown the optimality of debt contracts under similar constraints. I also derived a sufficient condition for validating the FOA. As is consistent with Chade and Swinkels (2020), the condition involves the concavities (in proportionate terms) of outcome distribution and disutility of action, and it is implied by Rogerson (1985)'s seminal Convexity of Distribution Function Condition and convex disutility.

The dual utility representation of Yaari (1987) brings structural simplicity to the principal's optimization problem, which leads to the bang-bang property (slope is either 0 or 1) of solutions. This simplicity should facilitate further analysis of moral hazard problems in more complex or applied environments. Applying the present framework to a broader range of economic settings is a promising direction for future research.

Appendix

A Debt and capped bonus contract

Appendix B shows that the solution candidates to (P) belong to a simple class of contracts, which I dub *debt and capped bonus contract*. In this section, I define this class and derive its property to simplify the expositions of Appendix B.

Recall that in (P) the principal is restricted to wage schedules that satisfy (FOA):

$$\int_{\underline{x}}^{\bar{x}} k[-F_a(x | a)] w'(x) dx = \psi'(a). \quad (\text{FOA})$$

By the implicit function theorem, there exists function $l: [\underline{x}, \bar{x}] \rightarrow [\underline{x}, \bar{x}]$ such that

$$\int_y^{l(y)} k[-F_a(x | a)] dx = \psi'(a).$$

Using this function l , I define debt and capped bonus contract w_{DB} as follows: for any $y \in [\underline{x}, \bar{y}]$, where $\bar{y} \equiv l^{-1}(\bar{x})$

$$w_{\text{DB}}(x | y, \underline{w}) = \begin{cases} \underline{w} & \text{if } x \in [\underline{x}, y] \\ \underline{w} + x - y & \text{if } x \in (y, l(y)] \\ \underline{w} + l(y) - y & \text{if } x \in [l(y), \bar{x}]. \end{cases}$$

That is, $w_{\text{DB}}(x | y, \underline{w})$ is the wage schedule such that (i) the base wage is $\underline{w}$; (ii) the agent receives “bonus” on $[y, l(y)]$; and (iii) the wage is constant outside $[y, l(y)]$. In particular, for any base wage $\underline{w}$, $w_{\text{DB}}(x | \underline{x}, \underline{w})$ is a *capped bonus contract*, and $w_{\text{DB}}(x | \bar{y}, \underline{w})$ is a *debt contract* (see Figure A.1).

For $w_{\text{DB}}(x | y, \underline{w})$, (PC) can be written as

$$\underline{w} \geq U_0 + \psi(a) - \int_y^{l(y)} k[1 - F(x | a)] dx \equiv B(y).$$

Note that under Strict MLRP, the implicit function theorem also implies $B'(y) > 0$.

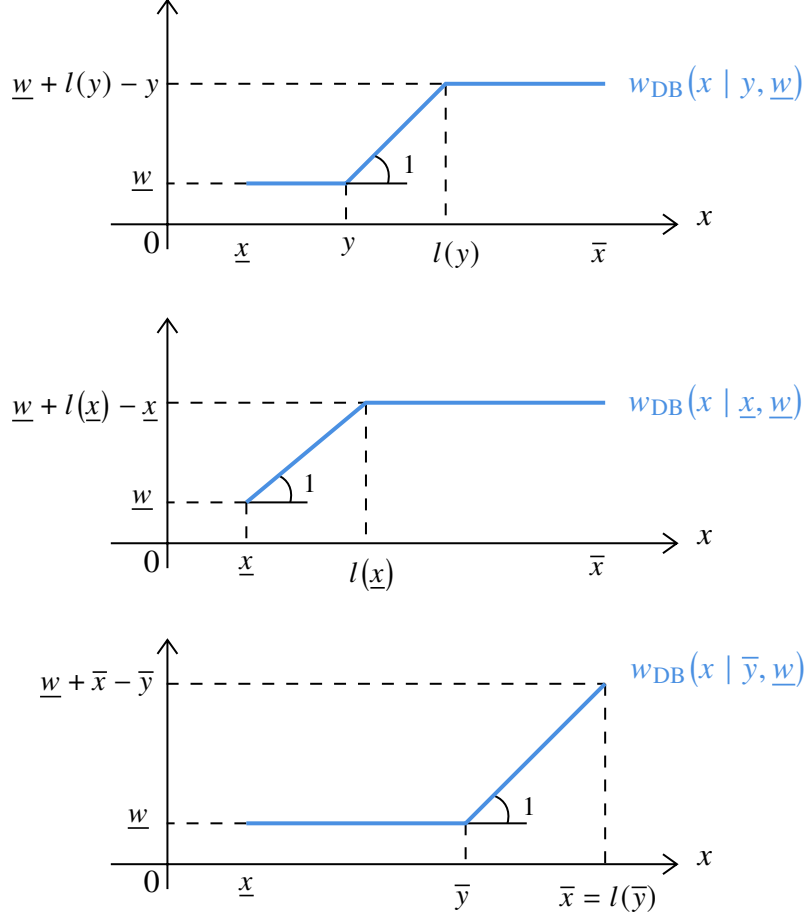


Figure A.1: Debt and capped bonus contracts. Debt contract and capped bonus contract are special cases of debt and capped bonus contract.

B Proof of Proposition 1

First, I show that Strict MLRP implies that $\frac{f}{1-F}$ is decreasing in action, the property repeatedly used in the proof of Proposition 1. Since $\frac{f_a}{f} = \frac{\partial}{\partial a} \ln f$, for any a and a' ,

$$\frac{f(x | a')}{f(x | a)} = \exp \left[\int_a^{a'} \frac{f_a(x | \tilde{a})}{f(x | \tilde{a})} d\tilde{a} \right].$$

Hence, by Strict MLRP, for any $x < x'$ and $a < a'$,

$$\frac{f(x' | a')}{f(x' | a)} > \frac{f(x | a')}{f(x | a)} \iff f(x' | a')f(x | a) > f(x | a')f(x' | a).$$

Integrating both sides with respect to x' from x to $\bar{x}$ and rearranging,

$$\frac{f(x | a)}{1 - F(x | a)} > \frac{f(x | a')}{1 - F(x | a')} \quad \text{for all } x \in [\underline{x}, \bar{x}]. \quad (\text{B.1})$$

This proves that $\frac{f}{1-F}$ is decreasing in a .

The proof of the proposition consists of two steps. First, using Pontryagin's maximum principle, I solve the relaxed problem (P) for fixed $w(\underline{x})$. Then, given the solution so obtained, I solve the optimization over $w(\underline{x})$.

B.1 Step 1: Solving the relaxed problem for fixed $w(\underline{x})$

Fix $w(\underline{x}) = \underline{w} \geq 0$. Define the Hamiltonian for (P) as follows:

$$H(w, u, p_1, p_2, p_3, x) = -p_0 w f + p_1 u + p_2 k(1 - F)u + p_3 k(-F_a)u.$$

Let $u^*(x)$ be a piecewise continuous control defined on $[\underline{x}, \bar{x}]$ that solves problem (P) and let $w^*(x)$, $\alpha^*(x)$ and $\beta^*(x)$ be the associated optimal paths.⁵ By Theorem 2 of [Seierstad and Sydsaeter \(1986, chap. 2\)](#), there exist constants p_0 , p_2^* , and p_3^* and a continuous and piecewise continuously differentiable adjoint function $p_1(x)$ such that, for all $x \in [\underline{x}, \bar{x}]$,

$$(p_0, p_1(x), p_2^*, p_3^*) \neq (0, 0, 0, 0) \quad (\text{B.2})$$

$$p_0 = 0 \text{ or } p_0 = 1 \quad (\text{B.3})$$

$$u^*(x) \text{ maximizes } H(w^*(x), u, p_1(x), p_2^*, p_3^*, x) \text{ for all } u \text{ in } [0, 1] \quad (\text{B.4})$$

$$\text{Except at the points of discontinuities of } u^*(x), p_1'(x) = p_0 f(x | a) \quad (\text{B.5})$$

$$p_1(\bar{x}) = 0, p_2^* \geq 0 \text{ and } p_2^*[\alpha^*(\bar{x}) - \psi(a) - U_0] = 0. \quad (\text{B.6})$$

Since the Hamiltonian is linear and hence concave in (w, u) , the maximized Hamiltonian $\widehat{H}(w, p_1, p_2, p_3, x) \equiv \max_{u \in [0, 1]} H(w, u, p_1, p_2, p_3, x)$ is also concave in w . Thus, by Theorem 5 of [Seierstad and Sydsaeter \(1986, chap. 2\)](#), the necessary conditions (B.2) to (B.6) are also

⁵I choose the value of $u(x)$ at a point of discontinuity x' as the left-hand limit of $u(x)$ at x' . I also assume that $u(x)$ is continuous at $\underline{x}$ and $\bar{x}$.

sufficient for optimality if they are satisfied with $p_0 = 1$.

By (B.5) and (B.6), $p_1(x) = -p_0[1 - F(x | a)]$. Substituting this into the Hamiltonian,

$$H(w, u, p_1(x), p_2^*, p_3^*, x) = -p_0 w f - p_0(1 - F)u + p_2 k(1 - F)u + p_3 k(-F_a)u,$$

and hence

$$\begin{aligned} H_u^*(x) &\equiv \frac{\partial H}{\partial u}(w^*(x), u, p_1(x), p_2^*, p_3^*, x) \\ &= (p_2^* k - p_0)[1 - F(x | a)] + p_3^* k[-F_a(x | a)]. \end{aligned}$$

A necessary condition for (B.4) is

$$u^*(x) = \begin{cases} 1 & \text{if } H_u^*(x) > 0 \\ 0 & \text{if } H_u^*(x) < 0. \end{cases} \quad (\text{B.7})$$

Suppose $p_0 = 0$. Then, $p_1(x) = 0$ for all x . Hence, by (B.2) and (B.6), at least one of $p_2^* > 0$ and $p_3^* \neq 0$ must hold.

Case 1-1: Suppose $p_2^* > 0$ and $p_3^* = 0$. Then, $H_u^*(x) = p_2^* k[1 - F(x | a)] > 0$ for all $x \in [\underline{x}, \bar{x}]$.

Hence, (B.7) implies $u^*(x) = 1$ for all x , which violates $\beta(\bar{x}) = \psi'(a)$ or equivalently (FOA).

Case 1-2: Suppose $p_2^* = 0$ and $p_3^* \neq 0$. Since $H_u^*(x) = p_3^* k[-F_a(x | a)]$, either $H_u^*(x) > 0$ or $H_u^*(x) < 0$ for all $x \in (\underline{x}, \bar{x})$. Hence, (B.7) implies $u^*(x) = 1$ or $u^*(x) = 0$ for all x , which violates (FOA).

Case 1-3: Suppose $p_2^* > 0$ and $p_3^* \neq 0$. Since, $H_u^*(x) = p_2^* k[1 - F(x | a)] + p_3^* k[-F_a(x | a)]$, $H_u^*(x) > 0 \Leftrightarrow p_3^* > p_2^* \left[\frac{1 - F(x|a)}{F_a(x|a)} \right] \equiv r(x)$. Note that (B.1) implies $r'(x) > 0$ and that $\lim_{x \rightarrow \underline{x}} r(x) = -\infty$. Hence, the unique candidate is a capped bonus contract:

$$w_{\text{DB}}(x | \underline{x}, \underline{w}) = \begin{cases} \underline{w} + x - \underline{x} & \text{if } x \in [\underline{x}, l(\underline{x})] \\ \underline{w} + l(\underline{x}) - \underline{x} & \text{if } x \in (l(\underline{x}), \bar{x}]. \end{cases}$$

By (B.6), I must have $\alpha^*(\bar{x}) - \psi(a) - U_0 = 0$ or equivalently binding (PC):

$$\underline{w} = B(\underline{x}).$$

Now suppose $p_0 = 1$. Then, $p_1(x) = -[1 - F(x | a)]$.

Case 2-1: Suppose $p_2^* = p_3^* = 0$. Then, $H_u^*(x) = -[1 - F(x | a)] < 0$ for all $x \in [\underline{x}, \bar{x}]$. Hence, (B.7) implies $u^*(x) = 1$ for all x , which violates (FOA).

Case 2-2: Suppose $p_2^* > 0$ and $p_3^* = 0$. Then, $H_u^*(x) = (p_2^*k - 1)[1 - F(x | a)]$. If $p_2^* \neq 1/k$, then (B.7) implies either $u^*(x) = 1$ or $u^*(x) = 0$ for all x , which violates (FOA). If $p_2^* = 1/k$, then any u^* is compatible with (B.7). In that case, (PC) must bind by (B.6).

In particular, the debt and capped bonus contract $w_{DB}(x | y, \underline{w})$ satisfy all the necessary conditions if

$$\underline{w} = B(y).$$

Case 2-3: Suppose $p_2^* = 0$ and $p_3^* \neq 0$. Since $H_u^*(x) = -[1 - F(x | a)] + p_3^*k[-F_a(x | a)]$, $H_u^*(x) > 0 \Leftrightarrow p_3^* > \frac{1-F(x|a)}{k[-F_a(x|a)]} \equiv s(x)$. Note that (B.1) implies $s'(x) < 0$ and that $\lim_{x \rightarrow \underline{x}} s(x) = +\infty$. Hence, the unique candidate is a debt contract:

$$w_{DB}(x | \bar{y}, \underline{w}) = \begin{cases} \underline{w} & \text{if } x \in [\underline{x}, \bar{y}] \\ \underline{w} + x - \bar{y} & \text{if } x \in (\bar{y}, \bar{x}], \end{cases}$$

and

$$\underline{w} \geq B(\bar{y}).$$

Case 2-4: Suppose $p_2^* > 0$ and $p_3^* \neq 0$. Since, $H_u^*(x) = (p_2^*k - 1)[1 - F(x | a)] + p_3^*k[-F_a(x | a)]$, $H_u^*(x) > 0 \Leftrightarrow p_3^* > \frac{(1-p_2^*k)[1-F(x|a)]}{k[-F_a(x|a)]} \equiv t(x)$.

- If $p_2^* = 1/k$, then $t(x) = 0$ for all x . Hence, (B.7) implies $u^*(x) = 1$ or $u^*(x) = 0$ for all x , which violates (FOA).
- If $p_2^* < 1/k$, then $\lim_{x \rightarrow \underline{x}} t(x) = +\infty$ and (B.1) implies $t'(x) < 0$. Hence, the unique

candidate is a debt contract $w_{\text{DB}}(x \mid \bar{y}, \underline{w})$ with binding (PC):

$$\underline{w} = B(\bar{y}).$$

- If $p_2^* > 1/k$, then $\lim_{x \rightarrow \underline{x}} t(x) = -\infty$ and (B.1) implies $t'(x) > 0$. Hence, the unique candidate is a capped bonus contract $w_{\text{DB}}(x \mid \underline{x}, \underline{w})$ with binding (PC):

$$\underline{w} = B(\underline{x}).$$

Thus, when $w(\underline{x})$ is fixed to $\underline{w}$, the optimal path w^* associated with the solution to (P) is given by

$$w^* = \begin{cases} w_{\text{DB}}(x \mid \underline{x}, \underline{w}) & \text{if } \underline{w} = B(\underline{x}) \\ w_{\text{DB}}(x \mid y, \underline{w}) & \text{if } \underline{w} = B(y) \\ w_{\text{DB}}(x \mid \bar{y}, \underline{w}) & \text{if } \underline{w} \geq B(\bar{y}). \end{cases} \quad (\text{B.8})$$

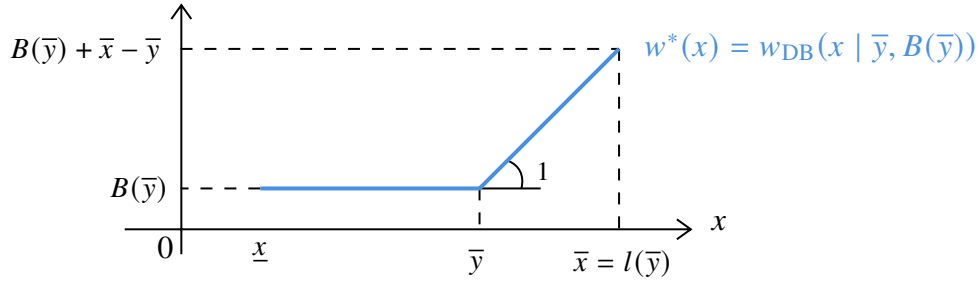
□

B.2 Step 2: Optimization over $w(\underline{x})$

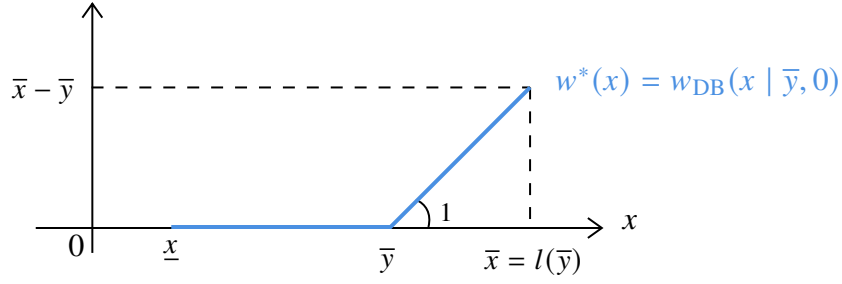
Given (B.8), I now solve the optimization over $\underline{w}$. Note that among the debt contracts $w_{\text{DB}}(x \mid \bar{y}, \underline{w})$, the one with $\underline{w} = B(\bar{y})$ minimizes the expected payment to the agent. Hence, it is without loss to focus on $w_{\text{DB}}(x \mid y, B(y))$. The principal's expected payment from this contract is

$$\begin{aligned} & \int_{\underline{x}}^y B(y) f(x \mid a) dx + \int_y^{l(y)} [B(y) + x - y] f(x \mid a) dx + \int_{l(y)}^{\bar{x}} [B(y) + l(y) - y] f(x \mid a) dx \\ &= U_0 + \psi(a) - \int_y^{l(y)} k[1 - F(x \mid a)] dx + \int_y^{l(y)} x f(x \mid a) dx - y[1 - F(y \mid a)] + l(y)[1 - F(l(y) \mid a)] \\ &= U_0 + \psi(a) + (1 - k) \int_y^{l(y)} [1 - F(x \mid a)] dx, \end{aligned}$$

where the second inequality follows from applying integration by parts to the fourth term in the second line. The implicit function theorem implies that the last term in the last line is decreasing in y . Thus, the principal always chooses debt contracts ($y = \bar{y}$). The optimal base wage is



(i) When $B(\bar{y}) > 0$



(ii) When $B(\bar{y}) \leq 0$

Figure B.1: The cost-minimizing wage schedule is always a debt contract. The base wage is zero when the limited liability constraint binds.

determined by $\underline{w}^* = \max\{B(\bar{y}), 0\}$. In summary, the solution w^* to the relaxed problem (P) is

$$w^* = \begin{cases} w_{\text{DB}}(x \mid \bar{y}, B(\bar{y})) & \text{if } B(\bar{y}) \geq 0 \\ w_{\text{DB}}(x \mid \bar{y}, 0) & \text{if } B(\bar{y}) < 0. \end{cases}$$

The cost-minimizing wage schedule w^* is depicted in Figure B.1.

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